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Squirrel App Collects Native Seeds and Supports Reforestation Using AWS Services

Wilder Climate Solutions and Accenture developed the Squirrel app to connect seed collectors with tree growers

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2 billion
trees

Improved
seed inventory data and traceability

10–100X
faster rate of climate change than fore



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Overview

Over the next decade, Canada aims to plant 2 billion trees. And by 2030, the United Nations has a goal to plant 1 trillion trees globally. As our planet warms due to climate change, planting more trees is one of the most efficient

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and cost-effective methods for carbon sequestration. However, it can be challenging for nurseries and researchers to find high quality, locally adapted native seeds that will successfully grow into biodiverse forests.

The Squirrel app, developed through a collaboration between Wilder Climate Solutions, Accenture, and Amazon Web Services (AWS), was created to address this challenge by connecting seed collectors, restoration ecologists, and tree growers. Using AWS, Squirrel allows restoration ecologists—sometimes working to replant forests after a fire—to request specific seeds. Then seed collectors can find, collect, catalog, geo-tag, and fulfill these requests.

Squirrel will help service the market for native seeds, improve their traceability —especially the crucial data about source—and enable smarter reforestation. All of this will come at a time when trees are a critical resource to fighting climate change and biodiversity loss.



Opportunity | Cultivating Connections Between Seed Collectors and Tree Growers

During the pandemic, Blaine Pearson, CEO and co-founder of Wilder Climate Solutions, became a seed collector. As she ventured into the forests near Toronto she learned to gather seed from a variety of locally adapted native trees and shrubs. With a freezer full of acorns and a continually growing collection, she began connecting with nurseries that might have demand for the seed.

After working on a national survey and report about the native seed supply in Canada, Pearson realized there were a number of major gaps hindering the market. At the beginning of the supply chain there is a lack of trained people doing the work of growing and collecting seeds. There is also a gap around market matching to bring together nurseries and collectors. Finally, collecting seeds is a rare skill. It tends to be learned over years of trial and error, with collectors using paper or rudimentary spreadsheets to catalog their collections—or nothing at all. The absence of verified and secure data leaves room for error and inhibits visibility that would help improve decision-making.

“Most seed collectors are still using analog forms of data capture, like pens and paper,” Pearson said. “That’s worked for a long time, but it leaves a lot on the table in terms of data visibility and traceability, which can enable better decision-making down the line. We wanted to be able to use technology to onboard a new generation of collectors, improve the data they collect, expedite the movement of demand signals through the supply chain, and enhance how data is connected to a seed as it moves through the system.”

Solution | Building an App so a Seed-Exchange Marketplace Can Take Root

After searching for a technology services company that shared their curiosity for the seed-to-tree supply chain, the team at Wilder Climate Solutions found Accenture. Together, the two organizations envisioned an app that would accomplish three goals: connect collectors and growers, digitize the process of seed collecting, and train novice seed collectors on best practices.

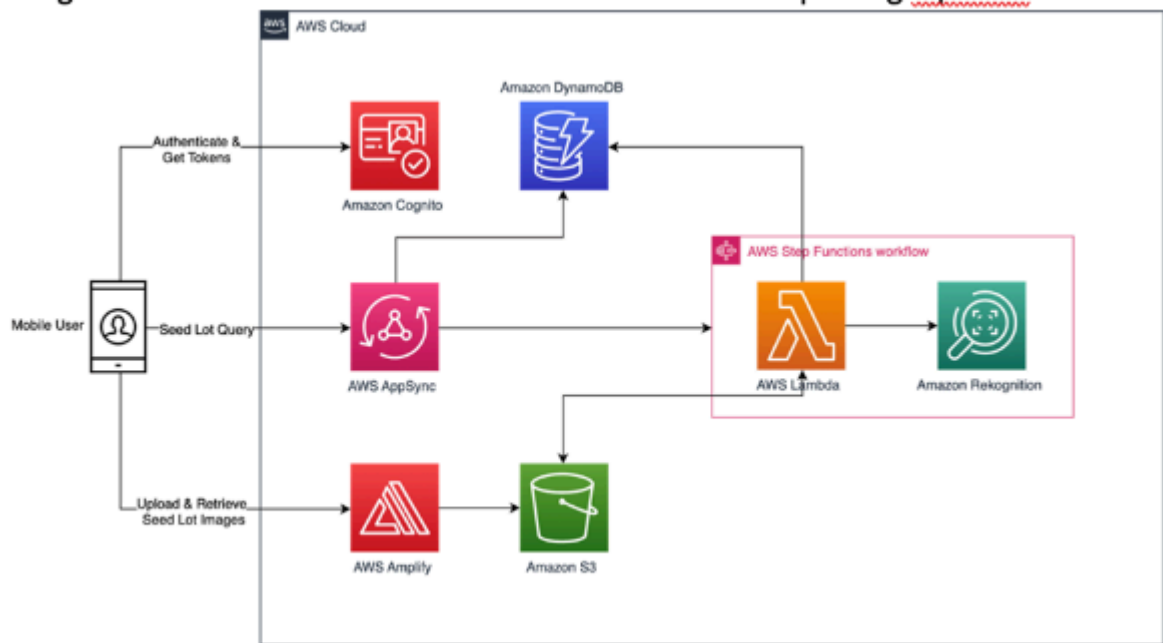
To turn this vision into the Squirrel app, Accenture turned to AWS. For several years, Accenture and AWS have worked on sustainability projects that help communities improve climate and biodiversity resilience.

“Accenture has a history of developing nature-based solutions on AWS, so we knew AWS would provide the best cloud services for us to create Squirrel,” said Julian Caspari, co-lead of the sustainability practice at Accenture.

Squirrel functions as a nearly infinite seed catalog for collectors to track the origin and viability of specimens they find. For new users, the app prompts them to take photos of where they found their seeds, as well as the seed itself. It also geotags their location and asks a series of questions about how they found the seed—was it on the ground, was the tree where it came from healthy, was it part of a tree stand? For each collection, Squirrel creates a QR code that serves as a unique tracker for those seeds throughout their entire lifecycle. As the seeds are planted and grow into trees, that QR code will still be able to show where the seeds originated and under what conditions.

All images uploaded into Squirrel are saved in an [Amazon Simple Storage Service](#) (Amazon S3) bucket and information about each seed is logged in a backend database powered by [Amazon DynamoDB](#). Users can also search Squirrel for particular seeds and [Amazon Rekognition](#) kicks in to analyze saved images and surface relevant photos and results. To build out the frontend of the app, the team at Accenture used [AWS Amplify](#).

Architecture Map



Outcome | Planting More Trees with Better Data

As usage of Squirrel grows, it will help seed collectors improve the data associated with their inventories. Having this information readily available in an app will not only streamline the process for collectors, but also help improve the traceability of seeds. “As a seed becomes a seedling, then moves around a greenhouse, then gets planted, the QR codes from Squirrel make it possible to trace it back to its original mother tree,” Pearson said. “That information enables research and emissions tracking, because we’ll know where a seed came from and can trace its entire lifecycle.”

For growers and others involved in ecological restoration work, Squirrel is a go-to marketplace to find and purchase seeds. “We’ve had really excellent uptake and reception because many tree growers are feeling the pressure to develop a greater supply of trees,” Pearson said. “They’re excited to have a resource that can help them get the right trees into production, faster.”

As climate change accelerates and takes a greater toll on ecosystems, trees will play a bigger role in fighting the effects and boosting resiliency. However, according to the 2020 State of Canada’s Forest Report, the rate of climate change is expected to outpace the ability of forests to adapt naturally—by 10 to 100X. “Everybody recognizes that trees are a vital part of our current toolkit in fighting climate change,” Pearson said. “As we lose more trees to wildfires, it becomes increasingly imperative to replant them. With Squirrel and AWS, we’re helping to improve access to seeds to expedite reforestation through better data and technology.”

About Wilder Climate Solutions

Wilder Climate Solutions leverages the power of data to fast-track global reforestation by producing nature-based climate solutions that improve biodiversity and enhance resiliency. The organization works with nature and technology to deliver climate-saving interventions that are based on science and focused on driving big results.

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Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL, fully managed database service with single-digit millisecond response times at any scale, enabling you to develop and run modern applications while only paying for what you use.

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